

NDP3® Assessment guidance

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Assessment of a child, presenting with a speech disorder, begins with a detailed case history, and consideration of his overall communication skills and development. Areas investigated may include: play, social skills, communication skills, verbal comprehension, expressive language, oral examination, phonology, articulation, voice, prosody, auditory skills, gross and fine motor skills, visual/tactile skills, non-verbal skills and emotional well-being (Clinical Guidelines by Consensus for Speech and Language Therapists, 1998). In many cases, other professionals, such as paediatricians, psychologists, occupational therapists, audiologists, will contribute to the assessment.

Initial assessments will suggest the nature and cause of the child's difficulties, and may indicate the presence of developmental verbal dyspraxia (See chapter 1 for discussion of characteristics of developmental verbal dyspraxia).

However, detailed investigation of the child's speech is necessary in order to plan intervention. Therapists typically employ a range of formal and informal assessment tools, to elicit single words, connected speech and a phonetic inventory, depending on the level of the child's speech and expressive language. Phonological analysis of this data reveals his sound system, at various positions in syllable structure, and relationships between the adult target and the child's production.

The Nuffield Dyspraxia Programme Assessment takes this process a stage further, recognising the dyspraxic child's increasing difficulty with longer words and phrases, by sampling and analysing separately words and sentences of increasing phonotactic complexity. Vowels are sampled systematically, as well as consonants. Evaluation of oral motor skills, diadochokinetic skills (DDK), and connected speech are also included. Closely related to the treatment programme, the Nuffield Dyspraxia Programme Assessment also provides a sensitive measure of progress.

Psycholinguistic approach

Since the Programme was last published in 1992, understanding of complex speech disorders has benefited from the development of psycholinguistic models of speech processing. Psycholinguistic models attempt to explain the process of speech, and therefore speech difficulties, as opposed to linguistic analysis, which describes the end result. Therapists at the Nuffield Centre have particularly been working with the model developed by Stackhouse and Wells (1997), and the discussion and diagrams below are based on this.

The basic tenet of a psycholinguistic approach is that speech processing involves the routing of information between ear, brain and mouth.

The first distinction to be made is whether information is being routed from the ear up to the brain (input processing), or from the brain down to the mouth (output processing). These can be thought of as two information channels: on the input side, speech information is received and decoded, and on the output side, speech information is encoded and transmitted.

In addition to the two information channels, the psycholinguistic approach assumes that there must be a store of information (a representation), which serves as a basis for recognising speech as well as generating speech output. The representation would include details of a word's meaning (semantic representation), its sound structure

(phonological representation), and instructions for articulation (motor programme). An important clinical distinction to make, is whether it is the stored information (lexical representation) which is faulty, or on-line processes, such as discrimination and motor programming. Assessment tasks must therefore be analysed to ascertain whether lexical representations must be accessed, may be accessed, or cannot be used, to be performed successfully.

It should also be noted that information may be routed in a top-down direction (utilising stored information) and/or a bottom-up direction (utilising peripheral sensory input) in order to perform a task. This applies to either side of the model. For example, as incoming speech is being processed (bottom-up), the brain may predict a part of it (e.g. the final syllable of "caterpi..." caterpillar). Equally the basically top-down processing of the output side of the model may be influenced by kinaesthetic feedback from the articulators. This is an important consideration in the design of therapy programmes, where it may be possible to utilise stronger processes to support weaker ones, and it may be necessary to experiment with various processes and processing routes in order to elicit a target.

When applied to clinical assessment, it is clear that children with speech disorders may have difficulty at one or more points along this processing route. Assessment tasks can be selected to check specific processes. In this way a profile of the child's processing strengths and weaknesses can be created, avoiding the need for diagnostic categorisation, where this is unhelpful, and facilitating well-targeted treatment.

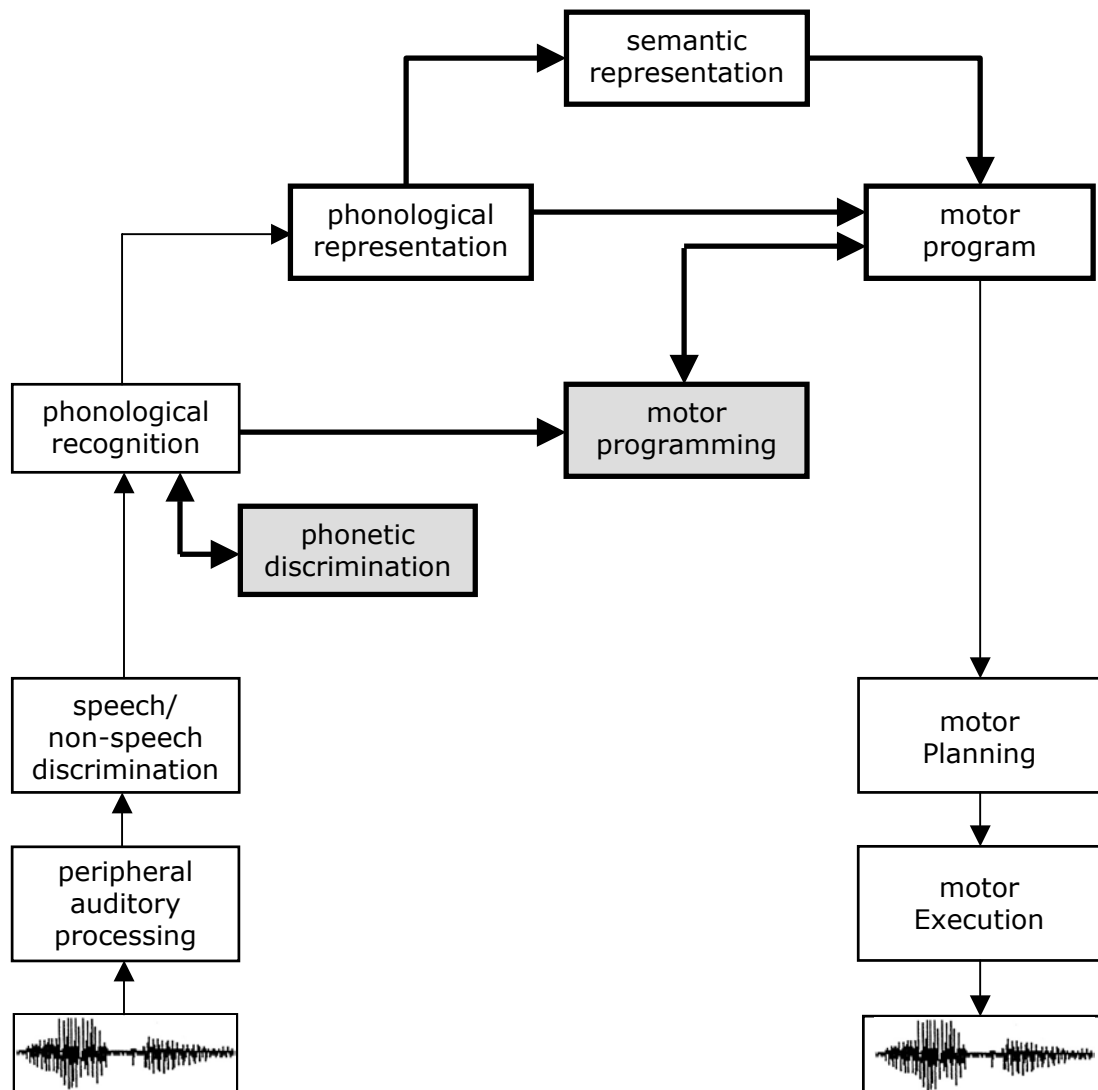
Psycholinguistic Model

The Stackhouse and Wells speech processing model is reproduced below, with permission. In the notes on each level of processing, which follow, we have combined the speech processing model with the questions from Stackhouse and Wells' speech processing profile. However, it should be noted that the questions do not strictly map onto particular processing levels, but relate to processing routes, (including or bypassing particular levels) required by types of assessment task. Two further questions (X, Y) have been added on the output side of the model, to clarify issues particularly relevant to children with features of developmental verbal dyspraxia.

Levels of processing are represented by boxes, and possible processing routes by arrows. Although not strictly hierarchical, there is a sense of processing order. For instance, the child must have discriminated the phonetic characteristics of a word before recognising it as a stored phonological representation. The levels do not differ in terms of absolute ease or difficulty, but in terms of relative cognitive sophistication. Off-line processing units appear as shaded boxes, indicating that processing at this level does not occur during the real time performance of the task i.e. on-line, but at some point afterwards i.e. off-line. Broad arrows indicate that information flows between boxes as part of a learning process, rather than in the on-line processing of familiar input.

For a detailed, and very readable, discussion of the speech processing model, its relation to normal speech acquisition and speech disorders, and analysis of assessment procedures, readers are recommended to refer to the original text (Stackhouse and Wells, 1997).

Speech Processing Model



© Whurr Publishers Ltd. From Stackhouse and Wells (1997): Children's Speech and Literacy Difficulties

Input processing

Peripheral auditory processing

The child must first hear the incoming speech signal, before any further decoding can take place. Hearing should be assessed as a prerequisite to any assessment of speech or language.

(A: Does the child have adequate auditory perception?)

Speech/nonspeech discrimination

Having perceived the signal, he must then recognise it as speech.

Phonological recognition

The phonological recognition level includes the recognition of the speech signal as a familiar language, and its segmentation first into words, and then into phonological units. Discrimination is thought to involve articulatory rehearsal. This may account for discrimination difficulties in children with difficulties on the output side of the model.

Phonological recognition, without reference to the child's stored representations, is tested by same/different judgements on nonwords (e.g. foss/fozz). More complex words, including multisyllabics and clusters should be used when testing older children.

(B: Can the child discriminate speech sounds without reference to lexical representations?)

Stackhouse and Wells included in their assessment framework a question specifically about the child's ability to discriminate legal/illegal non-words in a particular language. They suggest that this may be important in assessing bilingual children, and those with severe and persisting speech processing difficulties.

(C: Does the child have language specific representations of word structures?)

Phonological recognition may also be assessed by real word discrimination tasks, involving same/different judgements. However, these may be achieved by accessing stored phonological representations (see section under Lexical representations), avoiding the need to segment the signal. Again the child's age should be taken into account when selecting appropriate materials. Auditory memory loading can be high for these tasks (e.g. selecting odd one out from 3-4 spoken words) and this should also be considered when selecting tests.

(D: Can the child discriminate between real words?)

Phonetic discrimination

When the input contains novel phonetic material e.g. from an unfamiliar accent or language, this level of processing is required. Phonetic discrimination is also required for routine therapy tasks e.g. asking the child to discriminate at a phonetic level between sounds he substitutes e.g. [s] vs. [ʃ] – "did you hear a hissing sound or a slushy sound?"

Lexical representations (stored information)

Stored lexical representations for words include information on semantics, grammar and eventually the written form of the word, as well as the phonological representation and motor programme.

Phonological representation

The phonological representation includes the information necessary to identify the word, as produced by a range of speakers. Information is probably stored hierarchically, in terms of syllable, onset + rime, nucleus + coda, with phonemes specified by distinctive features.

In early speech development, representations would be specified only in sketchy terms (e.g. the /t/ in "tea" might be specified only as {+lingual, +plosive}), until the addition of "key" to the child's receptive vocabulary required that the term {+ alveolar} was added. Phonological representations must therefore be continually up-dated as the child extends his speech and language skills.

The accuracy of phonological representations can be checked, using real word discrimination tasks involving picture stimuli (i.e., asking the child to point to the picture named: e.g. pear/bear), or by right/wrong judgement tasks (e.g. "Is this a nake?", when looking at a picture of a snake). Of course, difficulties with peripheral auditory processing or phonological recognition may also impair performance on these tasks.

(E: Are the child's phonological representations accurate?)

The accuracy of phonological representations can also be checked without the spoken presentation of the item. In this case, the child is presented with picture items and asked to carry out a sorting task silently e.g. by onset, by coda etc.

(F: Is the child aware of the internal structure of phonological representations?)

Motor program

The motor program contains a series of gestural targets for the articulators, designed to achieve an acceptable pronunciation of the word. For example, accurate production of the word "tea" would require the following instructions: initiate airstream, close velopharyngeal sphincter, make closure with margins of tongue/alveolar ridge, release closure, assume raised front tongue body posture, spread lips and initiate voice.

Spontaneous speech and picture naming are used to assess accuracy of motor programs, but it should be remembered that these activities are also dependent on motor planning and execution.

(G: Can the child access accurate motor programs?)

Output processing

Motor programming

The motor programming level is thought to include a store of phonological units, probably onsets and rimes rather than individual phonemes, which can be combined to create new motor programs for unfamiliar words.

Motor programming is involved in early sound production and babble, and is crucial in early speech development, in creating and revising motor programs for a rapidly increasing vocabulary. Motor programming continues to be important, after the basics of speech are established, in dealing with new words, (e.g. for curriculum areas and hobbies, and the names of people and places). The manipulation of sounds is also involved in the development of alphabetic reading and spelling skills.

Motor programming can be tested by nonword repetition (although phonological recognition is also involved in these tasks), by rhyme production and in older children, by spoonerisms.

(X: Can the child create new motor programs?)

(H: Can the child manipulate phonological units?)

Motor planning

Once a motor pattern for a word is retrieved, or a new one created, a final plan for the utterance is formed, assembling gestural targets in the correct sequence, and taking account of the intended grammatical structure, phonetic environment and prosodic features. This is the level at which “slips of the tongue” occur in normal speakers. In children with developmental verbal dyspraxia, inconsistent realisation of targets occurs at this level, and there may be marked breakdown of articulation/phonology in connected speech.

Motor planning is assessed by comparing production of single words with production in sentences and/or connected speech, and by observation of inconsistent realisation of established motor programs. Motor planning is also assessed by DDK tasks, where the child is asked to repeat a target item on a number of occasions.

(Y: Can the child assemble a motor plan for an utterance effectively?)

Motor execution

Motor execution difficulties relate to problems with the structure or innervation of the articulators due, for example, to dysarthria or velopharyngeal insufficiency.

This level is assessed by means of an oral examination, observation of feeding skills, and DDK tasks. Consistent errors are observed in speech, which are often paralleled by difficulties with nonspeech functions.

(K: Does the child have adequate sound production skills?)

Self-monitoring

To complete the process, the child should be able to monitor his speech, identifying errors by comparison with stored phonological representations, and making an attempt to correct them.

(L: Does the child reject his own erroneous forms?)

Phonological awareness

Awareness of the phonological structure of words, and the ability to manipulate this at a conscious level, develops as the child matures. 3-4 year olds are able to demonstrate awareness of words and syllables, 4-5 year olds show awareness of onset and rime. Awareness of individual phonemes develops with exposure to letter sounds, at school (Goswami and Bryant 1990).

The child's developing awareness of the phonological structure of words is an important tool in therapy, enabling him to understand instructions such as "put a /t/ on the end".

Phonological awareness is assessed, as appropriate to the child's age, using both input and output tasks. Readers are recommended to refer to Stackhouse and Wells (1997), Chapter 3, for a detailed discussion of these issues.

Onset or rhyme matching, or "odd one out" discrimination tasks can be used to assess the child's awareness of the phonological structure of words. Tasks involving pictorial rather than spoken stimuli and responses ensure that phonological representations are used.

(F. Is the child aware of the internal structure of phonological representations?)

The ability to manipulate phonological representations can be tested by tasks such as production of rhyming words, segmentation, blending and deletion/addition tasks, (and in older children, spoonerisms). The use of nonword stimuli ensures that manipulation skills are tapped, and avoids the possibility that responses have been learned.

(H. Can the child manipulate phonological units?)

The Nuffield assessment from a psycholinguistic perspective

Stackhouse and Wells state that psycholinguistic assessment does not require new tests, but rather provides a framework for making decisions about what areas to assess and what materials/methods should be used.

The Nuffield Dyspraxia Programme Assessment is a test of output processing, providing information about the levels of motor programs, motor planning and motor execution. Levels targeted are shown below:

Pre-speech motor skills	Motor execution Motor programs (non-speech level)
DDK	Motor programming Motor planning Motor execution
Single sound imitation	Motor programs Motor programming Motor execution
Single word naming	Motor programs
Phrases and sentences	Motor planning
Connected speech	Motor planning with linguistic formulation

Skills are sampled and analysed separately at different levels of phonotactic complexity:

- single sounds
- CV and VC words
- CVCV words
- CVC words
- multisyllabic words
- consonant cluster words
- phrases and sentences

The assessment allows detailed analysis of what articulatory skills the child has, and what he is not yet able to do, and what phonological contrasts are maintained, at relevant levels of syllable structure.

Vowels are sampled systematically, as well as consonants. Evaluation of oral motor skills, DDK and connected speech are also included. Closely related to the treatment programme, the NDP Assessment also provides a sensitive measure of progress.

Most of the items are designed to be achieved by a 4 year old, but materials are also relevant to older children.

Psycholinguistic Profiling

It is recognised that while children with a possible diagnosis of developmental verbal dyspraxia are likely to have significant difficulties with one or more of the output components on the speech processing model, many also have inaccurate phonological representations, some discrimination difficulties, and poorly developed metaphonological skills.

In order to arrive at an overall profile of the child's speech processing, the Nuffield Dyspraxia Programme Assessment must therefore be supplemented by measures which assess other areas of the psycholinguistic model. A variety of test materials is available, including: The Auditory Discrimination and Attention test (MorganBarry 1989); Wepman's Auditory Discrimination Test (Wepman and Reynolds 1987); Aston Index (Newton and Thomson 1982); the Phonological Awareness Battery (Frederikson (ed) 1995); the Phonological Awareness test (Muter, Hulme and Snowling 1997); The Children's test of Nonword Repetition (Gathercole and Baddeley 1994). The selection of tests will depend on the child's age, and on the breadth and depth of assessment required.

Creating a microprofile

Although an individual child will have general patterns of strength and weakness in processing (e.g. phonological representations may be better than motor programs), difficulties in processing may vary between sound groups and levels of complexity (Corrin 2001). For example, a child who is unable to imitate either /ʃ/ or /k/, may have difficulty discriminating /s-ʃ/ but not /t-k/, if these are his typical substitutions. Similarly, he may have accurate phonological representations for shorter, but not for longer words, or motor execution difficulties affecting only tongue tip articulation.

Analysis of assessment data permits likely target areas for treatment to be identified, (for example: eliciting velar plosives, or establishing /p t k/ in CV structures). See Chapter 4: Treatment Planning for discussion. Target areas can then be examined in terms of the speech processing model, to identify strengths and weaknesses. These will determine the design of therapy tasks and the type of facilitatory techniques employed (see case studies).

Formal or informal probes may be carried out to tap different aspects of processing:

Auditory discrimination	Phonetic discrimination Phonological recognition Phonological representations
Imitation of real words to compare with naming	Motor programming, (probably without accessing phonological representations)
Non-word imitation	Motor programming
Wider vocabulary	Phonological representations Motor programs
Stimulability of missing items	Various routes: top-down/bottom up
Self monitoring	Feedback loop Phonological representations
Segmentation, matching	Awareness of the internal structure of phonological representations

NDP Assessment: subtests and speech probes

A description of each subtest, and the processing routes involved, is given below, followed by possible probes.

Oral motor assessment

This subtest assesses the motor execution level. Pure motor skills are elicited via non-verbal routes, at an automatic level, or by imitation. Additional information on motor skills in speech sound production is obtained from the single sound imitation section, which should be completed prior to administering the oro-motor assessment.

The oro-motor checklist is designed as a simple assessment for observing the child's functional skills, relating pre-speech oral skills to articulation of single sounds. The assessment is divided into 5 sections:

- Lips and jaw
- Tongue
- Airstream
- Voice
- Palate

It is not necessary to assess each of these areas in detail. Questions are posed at the beginning of each section to help advise whether assessment of a given area is advised.

An individual child's performance on oro-motor skills can be compared with normative information available (see Appendix 7 on normative issues: oral motor skills).

The assessment identifies motor skills underlying specific articulatory features, which may need to be developed in order to elicit specific sounds, and skills which the child already has, which might be used as a basis for extending his sound system. (See Appendix 4 for suggestions on eliciting specific sounds). For example:

- d) The child substitutes alveolars for labial sounds:
Does he achieve lip closure at an automatic level (feeding/controlling saliva)?
What is muscle tone like?
Can lip closure be achieved by imitation, instruction or with a physical prompt?
- e) Articulation of tongue tip sounds is imprecise:
What range of tongue tip movements are achieved at an automatic level (feeding, licking)?
What can be achieved with imitation, mirror, prompts?
What is muscle tone/appearance of the tongue like?
- f) Hypernasality/nasal emission is heard on single sounds:
Does the child achieve an oral airstream for blowing tasks?
Any abnormalities of muscle tone or structure?
Has the child been referred for palatal investigation?
- g) The child has a very restricted range of consonant and vowel sounds:
Investigate feeding skills and spontaneous sound-making.
What key movements/postures of the lips and tongue can be elicited by imitation or with prompts?
Observe structure of lips/tongue/palate, and muscle tone.
Observe wider motor skills.
- h) Voice quality and intonation are poor:
Is vocal tone better at an automatic level?
Can better voice be elicited with prompts and models?
Is there any sound or symbolic noise produced at a spontaneous level, which has better voice?
Is the child able to use basic pitch, volume and length contrasts?

The comments box should be used to note observations on oral structure, muscle tone, symmetry, involuntary movements, feeding, drooling, vocalisation, general motor skills.

Children with significant oro-motor difficulties may benefit from more detailed assessment of their feeding skills, and advice from an occupational therapist or physiotherapist on seating and techniques to normalise tone.

DDK

Syllable and word sequencing skills are assessed by imitation of a model once and then subsequently on five repetitions. For syllable sequences, the processing route is: auditory perception, phonological recognition, motor programming, motor planning and motor execution. For word sequences, following phonological recognition, the route is: motor program, motor planning and motor execution.

Six sequences are assessed involving two different types of stimuli: syllable sequences and real words (which can be supported with picture cues). Sequences involving two and three syllables are included. Two syllable sequences include two examples of sequences involving the same consonant in each syllable and two examples involving a change of consonant. Three syllable sequences involve a change of consonant in each of the three syllables. A practice example of each type of sequence is included to introduce the task. Each target is modelled once and the child is asked to imitate the model. He is then asked to repeat it 5 more times.

Scoring is based on accuracy, consistency and rate of production (see scoring section). Comparisons can be made between normal and atypical performance on such DDK tasks (see Appendix 7 on normative issues: DDK section-Williams and Stackhouse 2000).

Single sound inventory

The child's phonetic inventory is assessed by imitation. This requires the following processing route: auditory perception, phonetic discrimination, motor programming (or motor program, if the child has developed one during therapy!), motor planning and motor execution.

Consonants are divided on the assessment form into a basic range /p b m t d n k g f s h w j/ and harder sounds. It may be appropriate only to attempt the basic range with young children or those with very limited skills.

An individual child's performance on single sounds can be compared with normative information available (see Appendix 7: normative issues: speech sound emergence).

Vowels considered in terms of type (long/short/diphthong), and basic (i.e. those involving more extreme lip/tongue/jaw postures) as opposed to those requiring half-open jaw postures. It is recommended that all vowels are checked, as it is difficult to predict which will be easier for an individual child. Details of postures required for a standard set of English vowels are given in Appendix 3: Articulograms. Regional accents will require some modifications.

Probing speech processing

- Check consistency (motor planning), by attempting repetitions of a sound e.g. repeat "b" five times. Consonant sequences can be assessed through diadochokinetic tasks (see above). Vowel sequences can be assessed as appropriate e.g. can you say /i - u/, /a - i/ etc five times after me.
- Check stimulability of "missing" sounds, i.e. ways in which the motor program may be elicited. This involves experimenting with various processing routes: auditory models, visual cues (such as articulograms, lip reading and mirror), physical prompts (e.g. supported lip closure to elicit bilabials), semantic cues (e.g. food picture to help elicit /m/), and verbal instructions (e.g. "put your tongue on your teeth"). See also Appendix 4 for methods of eliciting sounds.
- Investigate underlying motor skills (motor programs and motor programming). Can the component movements/features of the target sound be elicited at a voluntary level? Do they occur in other sounds in the child's repertoire? Do they occur during a non-speech activity (feeding, crying, babble, animal noises, play, tooth-brushing)? If not, can more mature/wider range of movement patterns be facilitated? Liaise with Occupational therapist.
- Investigate discrimination of target from substituted sound(s).

Word Lists

Single words are elicited by picture naming, using the following processing route: semantic representation, motor program, motor planning, motor execution. Where the child is unable to name a picture, the word may be elicited by imitation. However, this should be noted, as the processing route now involves auditory discrimination and motor programming, rather than a stored motor program. For some children it may be very helpful to compare performance on naming tasks with that on repetition tasks and in such cases, the word lists can be presented separately for naming and repetition on different occasions.

20 words are elicited, through picture naming, at each of the following levels of syllable structure:

- CV and VC
- CVC
- CVCV (and CVCVC)
- Multisyllabics.

(Words of 3-4 syllables, with a range of syllable structures and stress patterns. Compound words and consonant clusters are also included.).

- Consonant clusters.

(Initial, medial and final clusters of various types are sampled.).

The pictures are organised into separate assessment flip-books by syllable structure.

With only 20 items per syllable structure, it is not possible to sample all sounds in all positions. The word lists have therefore been constructed to include as far as possible:

- *Vocabulary familiar to a 4 year old.*
- *A wide range of consonants and vowels in each list.*
- *Greater emphasis on the basic range of consonants at lower levels.*
- *Sequences of sounds which frequently cause difficulties for children with DVD.*

Children with developmental verbal dyspraxia have difficulty co-ordinating sequences of movement. This can result, at a phonological level, in syntagmatic processes, such as vowel or consonant assimilation. It may also result in more subtle articulatory/phonetic errors, where particular sounds/groups of sounds occur adjacent to one another. For example, when diphthongs are followed by nasals, the second element of the diphthong is often omitted. Sequences which may need specific teaching are listed on the assessment forms.

Data is analysed in terms of articulatory and phonological skills. Questions are included on the assessment forms as a guide. In depth assessment of speech processing, in relation to particular areas to be targeted, should be used as necessary to clarify the nature of the child's difficulties.

Probing speech processing

1 Check stimulability of target (i.e. ways of facilitating motor programming).

imitation of auditory model (e.g. boat)	Motor programming, attempting to by-pass phonological representations
imitation of split model (e.g. bow – t)	
imitation of non-word model (e.g. boit)	
Draw attention to target sound sequence (e.g. "It's got a /t/ at the end!")	Phonological awareness
Provide articulatory cues (verbal/visual) (e.g. "Put your tongue on your teeth"; use of mirror; articulogram)	Articulatory awareness and manipulation Motor execution

Check discrimination of substituted sounds

Minimal pairs: child selects picture to auditory model	Phonological recognition, routed via phonological representations
Same/different judgements (real words)	Ditto, but memory loading is higher
Same/different judgements (non words)	Phonological recognition, by-passing phonological representations

Check phonological representations

Right/wrong judgements of target word and errors, produced by therapist	Phonological representations
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Check motor programs for a wider group of words with target structure/sounds (e.g. wider range of CVC words with velar and alveolar plosives). Motor programming difficulties often mean that motor programs are not updated spontaneously.

Naming pictures	Motor programs
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Check motor programming

Imitating non-words	Motor programming
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Check motor planning (consistency of target)

Produce repetitions or sequences including target	Motor planning
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Check motor execution

Look for effects of limited range, speed, accuracy and strength of movement of lips, jaw, tongue, palate, respiration and larynx.	Motor execution: neuro-muscular control
Look for effects of structural problems (e.g. velo-pharyngeal insufficiency; dental anomalies; severe tongue tie)	Motor execution: structure

Phrases and sentences

Phrases and sentences are elicited by imitation, supported by pictures. The processing route includes input processing (auditory perception, phonological recognition) accessing lexical representations and motor programs for familiar words, followed by motor planning and motor execution. However, grammatical morphemes may be processed via a non-lexical route (discrimination and motor programming).

20 sentences are modelled, 4 - 12 syllables in length, and incorporating a range of sounds and syllable structures. This level of the assessment provides the opportunity to observe how the child deals with word joining, prosodic features and grammatical morphemes.

Organising utterances of increasing length represents a higher level of complexity for children with DVD. Breakdown is frequently seen at this level, despite adequate production of individual words. This may be due to difficulties at the motor planning stage of processing. Sentences include some words from the single word lists, in order to facilitate comparison.

Difficulties can also be seen at this level in children with greater impairment of motor execution, who have difficulty sustaining accurate articulation in longer utterances.

Because of the complexity of the task, however, difficulties may also result from problems with input processing, auditory memory and expressive language.

Noting strengths, weaknesses and strategies

The following features are noted specifically at phrase/sentence level:

- 1 Assimilation of consonants or vowels, simplifying the motor plan.

/ə taɪdə i? ɡɪɡɪn .../ - A tiger is digging...

Final consonant deletion, which may be the result of difficulty with word joining.

/ə kʌ? ə ti/ - A cup of tea.

Cluster reduction

/kætəpɪləz teɪndz ɪntu bʌtʌfaɪz/ - Caterpillars change into butterflies.

Imprecise vowels, probably due to motor execution difficulties.

/ni: maɪ? ə na:ʔha:θ/ - Lee made a lighthouse.

Imprecise consonants

/xɛt̪ɪ pɑːtɪ/ - Katy's party.

Syllable deletion

/daɪsə zɪjʌm/ - Dinosaur museum.

Junction. Various types of normal junction processes are modelled. A brief key is given here, but see Chapter 5: Connected speech, for more detailed information. (# is used to indicate a word boundary, and ' to mark syllables, where necessary).

- /j/ - A /j/ is inserted between 2 vowels, if the first ends in a high front/spread /i/ posture. For example, "Daddy is talking" is produced as /dædɪjɪz'tɔːkɪŋ/.

- C#V – Where a final consonant is followed by a vowel, the syllable structure of the sequence is reorganised so that these 2 sounds become a CV syllable. Thus “A cup of tea” is produced as /ə'kʌ'pʌv'ti/.
- C#C – Where a final plosive is followed by an initial plosive, the first is not released, and may be assimilated to the second. For example, “Two sad cats” may be produced as /tusæd'kæts/ or /tu sæg'kæts/. However, the second C#C sentence involves a fricative followed by a plosive, and both consonants should be heard: “I saw five camels...” /aɪsəfaɪvkæm'lz.../.
- /n/ - “A” becomes “an” before a vowel, and syllable structure is reorganised. For example, “Get an apple” is produced as /'gɛ'tə'næ'pl/.
- /w/- A /w/ is inserted between 2 vowels if the first ends in a high back/rounded /u/ posture. For example, “Go in the house” is produced as /gəʊwɪnðəhaus/.
- /r/ - A /r/ is inserted between 2 vowels when the first vowel sound could be spelt with an “r” digraph (ar, er/ir/ur, or (aw). For example “Peter is eating his dinner” is produced as /'pi'tə'rɪ'zi'tɪŋ'hɪz'dɪ'nə/.

Rate of speech/stress patterns. Slow, syllabic speech, rapid, dysfluent or variable pace and rhythm should be noted. Although sometimes said to be an unfortunate consequence of therapy approaches such as the NDP, a slow regular pace can be a very useful strategy for some children to achieve intelligibility, despite very severe speech difficulties. Some discover this for themselves, so this possibility should be considered when a child presents with slow/syllabic speech.

Voice quality, pitch and volume problems should be noted, as at lower levels.

Nasality. Hyponasality and hypernasality/nasal emission should be noted.

Intonation. Note anything that sounds “odd”: monotone/very narrow pitch range, sing-song quality, only one intonation contour/nuclear tone, inappropriate intonation pattern (sounds like a question/statement/command!), wrong focus (stress on the wrong word). Does the child imitate your intonation readily? See also speech probes, below.

Compensatory strategies. Is the child doing anything that helps his intelligibility? For example, pacing his speech, producing one word at a time. Is he aware of doing it? It may still be that he should be helped to use a more effective strategy, or that he is doing something that doesn't help at all, such as exaggerating intonation.

Probing speech processing

Further assessment of intonation can be carried out by asking the child to imitate a sentence several times with different intonation (nuclear tones/focus). More information on junction and intonation can be found in Chapter 5: From sentence level to connected speech section.

Connected speech checklist

The connected speech checklist provides a framework for looking at the child's connected speech, monitoring progress, and determining what aspects of connected speech may be impairing intelligibility. The scoring system also produces a severity rating, which we feel is appropriate for children of 4-5 years upwards.

In processing terms, spontaneous connected speech involves linguistic formulation, accessing lexical representations and motor programs, motor planning of complex grammatical/articulatory/prosodic sequences.

The checklist is based on a sample of the child's typical (spontaneous) connected speech. It may be elicited by means of 1:1 conversation/play, or by a story retell task. The topic can be manipulated to ensure that age-appropriate challenges are included (e.g. complex vocabulary from a school project or hobby, explanation of an activity or concept, increasing linguistic loading).

The sample should be tape recorded, and analysed in the following ways:

Listen first at a broad level, noting

- Pace, rhythm, voice, resonance (see Phrase/Sentence level).
- Consistency.

There are often slower, better articulated sections and faster, less clear parts, as the child learns to pace and monitor his speech effectively. Alternatively variation may be associated with different linguistic demands.

- Any strategies the child appears to be using to improve intelligibility.

Then listen more specifically for features of phonology and articulation

- Range of syllable structures, and what happens to final/medial consonants, clusters and weak syllables.
- Range of consonants and vowels used.
- Maintenance of basic place, voice and manner contrasts.
- Precision of articulation.
- Production of grammatical morphemes. These are difficult because they usually have to be unstressed, but children are often unsure of the phonological structure they are trying to reduce.
- Junction: how the child deals with joining sounds at word boundaries (e.g. omit final consonants, articulate one word at a time, or are they developing appropriate junction processes? (see under Phrases and Sentences this chapter).

The 80% use criterion should be estimated rather than calculated formally, and reflects features which are generally present in the child's speech.

By totalling the score on the connected speech checklist, an approximate severity rating, for children of 5 years upwards, can be given:

0 – 7	severe
8 – 15	moderate
16 – 20	mild

By deducting the one item on the checklist which would not be expected of a 4 year old (full range of fricatives and affricates, apart from /θ ð/), the rating could also be applied to 4 year olds:

0 – 6	severe
7 – 15	moderate
16 – 19	mild

Intelligibility Rating

Having considered various methods of rating intelligibility, we have found the disability scale from the phonological disorders section of Therapy Outcome Measures (Enderby and John, 1997) to be the simplest and most relevant. This is reproduced below, with permission.

Identify descriptor that is “**best fit**”: The patient does not have to have each feature mentioned. Use 0.5 to indicate if patient is slightly better or worse than a descriptor.

- 0 Completely unintelligible to familiar and nonfamiliar listeners.
- 1 Partly intelligible to familiar listeners in known context, communication partner bears the burden of responsibility.
- 2 Intelligible to familiar listeners in context, partly intelligible in context with nonfamiliar listeners, single words clear connected speech poor.
- 3 Usually intelligible to familiar listeners in and out of context, variable intelligibility in context with nonfamiliar listeners /free spontaneous speech often unintelligible.
- 4 Minor problems but intelligible to everyone, occasionally loses intelligibility at times e.g. when excited or speaking against noise etc.
- 5 Intelligible at age-appropriate level to familiar listeners and nonfamiliar listeners.

Administration, analysis, scoring and interpretation

Top Tips

- Investigate articulatory skills and phonological system at each level of complexity
- Note problems with voice and prosody
- Identify speech skills to be targeted at particular levels in therapy
- Investigate underlying motor skills for possible causes of sounds not present
- Investigate the speech processing system as it affects skills to be targeted
- Investigate the stimulability of possible targets
- Investigate the consistency of skills already evident
- Assess phonological awareness, as appropriate to the child's age

Administration

The NDP assessment is intended to be used flexibly. There is no need to assess higher levels if the child is clearly going to be working at single sound and CV levels. However, it may be useful to try just a couple of items from harder sections, to enable comparison with subsequent assessments. The assessment may be carried out over several sessions or completed in a single session.

Instructions for administration and scoring are included on the assessment forms.

Oral motor assessment

This section should be administered after the speech assessment, as it utilises information from the single sound section. It is not necessary to complete all sections- questions are posed at the beginning of each section (lips and jaw, tongue, airstream, voice, palate) to help the clinician decide whether a particular area should be assessed. Instructions for tasks are given verbally and demonstrated. Comments should be recorded regarding both structure and function. This section is not scored as a whole.

Single sounds

Single sounds are elicited by imitation. It is recommended that articulatory features (e.g. dental placement of alveolars, hypernasality, weak closure, breathy voice) are noted, as these are often important factors in identifying the nature of the child's problems and need to be incorporated in treatment planning.

Diadochokinetic skills

The target is modelled once and the child is asked to imitate the sequence. He is then asked to repeat it five times. Picture cues may be used for the real word items (from the CVCV section of the Programme).

It is recommended that each production is transcribed phonetically on the scoring sheet. It may be necessary to tape record this section to ensure an accurate measure is made. Note inaccuracies, inconsistency and unusually fast/slow rates of production. Young children may need help to understand when they have produced 5 productions e.g. using a tick box chart or by the therapist holding up a finger each time a production is said. The time taken to produce 5 repetitions should be recorded using a stop watch.

Word lists

These are elicited by picture naming. Words which the child cannot name should be imitated, and noted accordingly (see under speech processing above).

It is recommended that responses are transcribed, and articulatory patterns noted, including suprasegmental features: vocal tone, pitch, volume, resonance.

Phrases and sentences

The phrases and sentences are elicited by imitation, while the child is looking at the appropriate picture. They should be modelled at a slow, normal pace (approximately 3-4 syllables per second), with normal intonation and junction between words (i.e. not extra clearly, with pauses/glottal stops between every word).

Specific junction strategies are marked in some phrases, and care should be taken to include these in the model, if they are to be assessed in the child's production.

It is recommended that the sentences are tape recorded for later transcription (see notes this chapter for features to observe).

Connected speech

A taped sample of the child's typical (spontaneous) connected speech is elicited, by conversation or a story retell task. The sample is analysed, (see description, this chapter), and the checklist completed. The score can be related to a severity rating scale (see this chapter) for children of 4-5 years upwards.

Analysis of data

Data are analysed as appropriate, at each level, in terms of:

- Sounds, or sequences of sounds, which are correct
- Sounds, or sequences of sounds, which are incorrect
- Patterns in incorrect sounds/sequences
- Problems with syllable structure and junction between sounds
- Voice and prosodic features

Questions are included on the assessment forms as a guide to analysis. Problem areas, which are identified as next treatment targets, can then be probed for processing strengths and weaknesses (see above). The results of these probes will guide the choice of therapy approach(es) for each target. See assessment/treatment planning exercise below.

Scoring

A simple scoring system is included, in order that data can be compared and presented in numerical form, if desired, or in the form of a graph:

Single sounds	1	=	local adult model
	0	=	any significant error
DDK	1	=	can repeat sequence accurately once
	1	=	can repeat sequence accurately five times
	1	=	can produce 5 identical repetitions, even if inaccurate
	1	=	can repeat 5 repetitions in 3 seconds (2 syllable items) or in 4 seconds (3 syllable items)

NB Only syllable initial consonant accuracy is scored e.g. in “patacake”, only accuracy of p-t-k is scored.

Single words:	1	=	local adult model
	0	=	any significant error
Phrases/sentences	2	=	local adult model
	1	=	less than specified number of errors (see assessment form for details)
	0	=	more than specified number of errors

A significant error would normally be one which potentially compromised intelligibility. This could either be one which neutralised a phonemic contrast (e.g. /s/ → /ʃ/), or one which reduced clarity of articulation to a significant extent (e.g. /s/ → /t/). Another way of looking at this is to decide whether the child’s production falls outside the adult phoneme boundary, and is unacceptable production. For example, a dental [ʃ] falls within the phoneme boundary, and would be scored as correct, whereas [t ʃ θ] would all be incorrect. Where several attempts are made at one item, the best attempt should be scored. Further examples of scoring can be found in the assessment exercise, at the end of Chapter 4: Treatment Planning.

Scoring systems never deal adequately with all situations. It may be appropriate to modify this system for particular purposes. However, where such modifications are made, this should be noted clearly.

Percentage scores can be calculated for each section of the assessment. Care should be taken in interpreting scores when all items in a section have not been attempted.

Scoring instructions for each section are included on the assessment form.

Interpretation

The NDP assessment is not a standardised tool. However, data collected from typically-developing 3 and 4 year old children (Thompson 2009; Walrawens 2010) can be used to interpret the performance of children with features of developmental verbal dyspraxia:

- 3 & 4 year olds scored 85% or above on: single sounds, CV/VC words, CVCV words; CVC words and cluster words.
- 4 year olds scored 80% and above on multisyllabic words, phrases and sentences.
- 3 year olds scored 70% and above on multisyllabic words, phrases and sentences.
- As might be expected, there was more individual variation amongst 3 year olds.

Notes on assessment of younger children - further suggestions:

(i.e.: children with very limited language, vocabulary, attention, cooperation on formal testing)

- transcribe words child uses
- get parent to list child's words and approximate articulation
- name familiar objects/toys, including a variety of syllable structures and sounds
- imitation of a basic range of consonants /p b m t d n k g f s h w j (ʃ) (l)/
- imitation of as many vowels as possible, but start with basics /i a u ai iə au/
- symbolic noises
- imitation of CV syllables in play
- attempt 2-3 words from CVCV, CVC, multisyllabic and cluster lists, if possible, for comparison with subsequent assessments
- observe oral skills and feeding, and attempt imitation of oral movement in play
- note use of voice
- bear in mind child's age/developmental level, when interpreting results

Notes on assessment of older children - further suggestions:

(i.e. children who cope well with most items on the word lists, but still have problems with intelligibility)

- Start by listening to connected speech, and assessing intelligibility
- Work backwards to discover reasons for breakdown in intelligibility
- Check wider range of words at each level
- Check harder consonants /ʃ tʃ dʒ v z ʒ θ ð r/
- Check imitation of complex unfamiliar words and nonwords
- Check vowels, especially half-open postures
- Check problematic sound sequences, shown on treatment records
- Check oro-motor and sequencing skills
- Check voice and airstream
- Check prosodics: intonation, stress, rate of speech, fluency
- Check syllable/word joining strategies
- Note compensatory strategies
- Bear in mind the influence of language difficulties
- Assess phonological awareness and manipulation

Assessment and treatment planning exercise

Background information

The data below comes from a 6; 9 year old girl (E). Initial assessment shows that she has wider language and communication difficulties, but age appropriate literacy skills (including reading/spelling of non-words). She has a history of severe oro-motor dyspraxia. Her hearing is normal.

Analyse the assessment data, using the questions on the assessment forms as a guide. After reading the Treatment Planning section below, identify next treatment targets for E, and note any probes you would wish to make. A solution is given at the end of Chapter 4: Treatment Planning.

Diacritics used in the transcription are as follows:

- Ç - dental articulation
- Ç ʋ - more open articulation
- Ç ʋ̥ - devoiced

Name E

Date _____

Age 6;9 years

Single Consonants & Vowels

Elicited by picture naming.

Scoring: 1 = correct (local adult model)
 0 = incorrect/unable to attempt
 NT = not tested

Target	Response	
Easier consonants	p	<i>p</i> 1
	b	<i>b</i> 1
	m	<i>m</i> 1
	t	<i>d</i> 0
	d	<i>d</i> 1
	n	<i>n</i> 1
	k	<i>t</i> 0
	g	<i>d</i> 0
	h	<i>h</i> 1
	w	<i>w</i> 1
	j	<i>j</i> 1
	f	<i>f</i> 1
	s	<i>θ</i> 0
Harder consonants	ʃ	Silent posture 0
	l	<i>ðə</i> 0
	ŋ	NT
	v	NT
	z	NT
	tʃ	<i>t</i> 0
	dʒ	<i>d</i> 0
	θ	NT
	ð	NT
	r	<i>ð</i> 0
	ʒ	NT

i	<i>i</i>	1
ɑ	<i>ɑ</i>	1
u	<i>u</i>	1
ɔ	<i>ʊ</i>	0
ɜ	<i>ɜ</i>	1
εə	<i>æ:</i>	0
aɪ	<i>aɪ</i>	1
ɑʊ	<i>ɑʊ</i>	1
ɪə	<i>ɪə</i>	1
əʊ	<i>əʊ</i>	1
ɔɪ	<i>ɔwɪ</i>	0
eɪ	<i>aɪ</i>	0
ɪ	<i>ɪ</i>	1
æ	<i>æ</i>	1
ʊ	<i>ʊ</i>	1
ɒ	<i>ʊ</i>	0
ʌ	<i>ʌ</i>	1
ε	<i>ʌ</i>	0

TOTAL consonants /24	(9)
TOTAL vowels /18	12
TOTAL /42	
%	

What consonants are imitated correctly?

What consonants are incorrect? Any patterns?

Can any be elicited in other ways/ occur spontaneously?

What vowels are imitated correctly?

What vowels are incorrect? Any patterns?

Can any be elicited in other ways or occur spontaneously?

Any voice features to comment on?
Slightly hoarse/breathy + low, static pitch

Name EDate _____ Age 6;9 years

CV & VC Words

Target	Response	
moo	<i>mu</i>	1
door	<i>tu</i>	0
bee	<i>bi</i>	1
go	<i>tau</i>	0
knee	<i>ni</i>	1
deer	<i>dɪə</i>	1
boy	<i>bɔɪ</i>	1
car	<i>ta:</i>	0
cow	<i>dau</i>	0
two	<i>du</i>	0
pay	<i>bəɪ</i>	0
four	<i>fʊ</i>	0
sea	<i>θi ti θi</i>	0
shoe	<i>fu</i>	0
chair	<i>θɛ teə</i>	0
lie	<i>ðai</i>	0
out	<i>au?t</i>	1
up	<i>ʌ? ʌ?bʌ ʌp ʌp</i>	1
arm	<i>a:m</i>	1
ice	<i>aɪθ</i>	0
	TOTAL /20	8
	%	40

Elicited by picture naming.

Scoring: 1 = correct (local adult model),
 0 = incorrect/unable to attempt
 NT = not tested

What consonants are used correctly?

What consonants are incorrect? Any patterns?

What vowels are used correctly?

What vowels are incorrect? Any patterns?

Are both CV and VC syllable structures used?

Are sounds joined smoothly?

Any voice/prosodic features to comment on?

Level pitch, sing-song quality

Name E

Date _____

Age 6;9 years

CVC Words		
Target	Response	
cake	<i>d̥aɪ?tθ</i>	0
boat	<i>bəʊ?tθ</i>	0
bird	<i>bɜ:d̥</i>	1
down	<i>d̥aʊn</i>	1
moon	<i>mu:n</i>	1
dog	<i>d̥ɒ?t</i>	0
girl	<i>d̥ɜ:n</i>	0
tap	<i>t̥æ?pφ</i>	0
pig	<i>b̥ɪ?d̥</i>	0
cat	<i>d̥æ?t̥</i>	0
duck	<i>d̥ʌ?t̥</i>	0
farm	<i>fɑ:m̥</i>	0
horse	<i>hɔ?θ</i>	0
leaf	<i>ð̥i?f</i>	0
sun	<i>fʌn</i>	0
fish	<i>fɪ?θ</i>	0
light	<i>ð̥aɪt̥</i>	0
watch	<i>wɒ?θ</i>	0
jam	<i>d̥æm</i>	0
web	NT	
	TOTAL /20	3
	%	

Elicited by picture naming.

Scoring: 1 = correct (local adult model),
 0 = incorrect/unable to attempt
 NT = not tested

What consonants does the child use in initial position?

What consonants does the child use in final position?

What processes are seen?

Stopping

Prevocalic voicing

Fronting

Gliding

Backing

Final consonant deletion

Assimilation

Metathesis

Glottal substitution

Favourite articulation

Inconsistency

Imprecise placement

Weak articulation

Inadequate airflow

What vowels are used?

long vowels

diphthongs

short vowels

Any processes affecting vowels?

Are final consonants joined smoothly?

Lengthening

Lack of half-open vowels

Shortening

Neutralisation

Reduction of diphthongs

Nasalisation

Any voice/prosodic features to comment on?

Slow, deliberate articulation

Name E

Date _____

Age 6;9 years

CVCV Words		
Target	Response	
mummy	<i>mami</i>	1
baby	<i>beibi</i>	1
teddy	<i>dædi</i>	0
cooker	<i>tu?tɪ</i>	0
dirty	<i>dʒ?di</i>	0
dinner	<i>dɪnə</i>	1
party	<i>bati</i>	0
table	<i>tɛi?bu</i>	0
picking	<i>bi?tin</i>	0
cowboy	<i>taubɔi</i>	0
kettle	<i>dætu</i>	0
garden	<i>dɑ:dʌn</i>	0
tiger	<i>taɪdə</i>	0
water	<i>wɪ?dʌ</i>	0
fire	<i>faijɪ</i>	0
jelly	<i>dædʒi</i>	0
kitchen	<i>dɪ?dʌn</i>	0
coffee	<i>tɒ?fi</i>	0
tissue	<i>dɪ?dʒu</i>	0
sitting	<i>fi?fɪm</i>	0
	TOTAL /20	3
	%	15

Elicited by picture naming.

Scoring: 1 = correct (local adult model),

0 = incorrect/unable to attempt

NT = not tested

Are both syllables marked?

What consonants does the child use in initial position?

What consonants does the child use in medial position?

What processes are seen?

Stopping

Prevocalic voicing

Fronting

Gliding

Backing

Imprecise placement

Weak articulation

Inadequate airflow

Omission

Assimilation

Metathesis

Glottal substitution

Favourite articulation

Inconsistency

What vowels are used?

long vowels

diphthongs

short vowels

Any processes affecting vowels?

Are syllables joined smoothly?

Lengthening

Lack of half-open vowels

Shortening

Neutralisation

Reduction of diphthongs

Nasalisation

Any voice/prosodic features to comment on?

Glottal insertion, substitution

Omission of sound, or part of

Deletion of weak syllable

Syllables often produced with equal stress

Name EDate _____ Age 6;9 years

Complex & Multisyllabic Words

Target	Response	
pocket	<i>bʊ?dæ̃t</i>	0
ticket	<i>d̃ɪ?dæ̃t</i>	0
chocolate	<i>d̃ɒ?dæ̃t</i>	0
banana	<i>bæ̃næ̃næ̃</i>	0
birthday cake	<i>bɜːd̃eɪ d̃eɪ?t</i>	0
dinosaur	<i>d̃aɪnəʊθə̃</i>	0
caterpillar	<i>t̃æt̃æpɪj̃l</i>	0
helicopter	<i>hæ̃jɪtə?t̃l</i>	0
aeroplane	<i>hau?peɪ̃n</i>	0
crocodile	<i>t̃ʊ t̃ʊ?daɪ̃v</i>	0
television	<i>θ̃ɛwət̃ɪt̃ʊn</i>	0
ambulance	<i>(refused)</i>	0
fire engine	<i>væ̃væ̃ væ̃vɪ̃n</i>	0
washing machine	<i>wɒ̃θɪ̃n mæ̃θɪ̃n</i>	0
computer	<i>t̃əʊp̃ʊt̃l</i>	0
butterfly	<i>b̃l̃t̃lθaɪ̃</i>	0
animals	<i>?æ̃nɪ..?æ̃nɪ..?æ̃nɪ</i>	0
umbrella	<i>ʌ̃m̃d̃ɛ̃ñl</i>	0
hospital	<i>p̃fɒ?tɪ?t̃l</i>	0
sausages	<i>f̃ɒ?f̃ɒ?hɪ?</i>	0
	TOTAL /20	0
	%	0

Elicited by picture naming.

Scoring: 1 = correct (local adult model)
 0 = incorrect/unable to attempt
 NT = not tested

How many syllables can the child signal correctly?

Are plosives & nasals included, with place, voice and manner contrasts maintained?

Are fricatives and affricates included appropriately?

Are clusters included, initially, medially & finally?

Any vowel errors/distortions?

Are syllables joined smoothly?

Are stress patterns appropriate?

Tends to produce one syllable at a time, with equal stress

Any voice/prosodic features to comment on?

Name EDate _____ Age 6;9 years

Consonant Cluster Words		
Target	Response	
star	<i>tʒ</i>	0
sky	<i>taɪ</i>	0
snake	<i>deɪ?t</i>	0
swing	<i>wɪn</i>	0
blue	<i>bu:</i>	0
flower	<i>fjaʊwə</i>	0
clock	<i>dɒt</i>	0
tree	<i>tri:</i>	0
pram	<i>tæm</i>	0
frog	<i>dɒt</i>	0
fast	<i>fa?θ</i>	0
boats	<i>bəʊt</i>	0
hand	<i>hæm</i>	0
crisps	<i>dɪ?p</i>	0
biscuit	<i>bɪ?dɪt</i>	0
lift	<i>vɪ?θ</i>	0
jump	<i>dʌmp</i>	0
monkey	<i>mʌn?dɪ</i>	0
ice-cream	<i>aɪ?ɪm</i>	0
Christmas	<i>(refused)</i>	0
	TOTAL	0
	%	0

Elicited by picture naming.

Scoring: 1 = correct (local adult model)
 0 = incorrect/unable to attempt
 NT = not tested

How does the child deal with initial clusters?

/s/ +C	C+ /l/	C+ /r/

How does he deal with medial clusters?

Nasal + C	/s/ + C	C + /t.d/

How does he deal with final clusters?

Nasal + C	/l/ + C	/s/ + C
C + /s.z/	C + /t.d/	

Are clusters articulated smoothly? Any intrusive [ʔ]
 or [ə]?

Any voice/prosodic features to comment on?

Name EDate _____ Age 6;9 years

Phrases & Sentences (1)

Elicited by imitation, supported by pictures. (Model at a slow normal rate, in normal conversational speech. Take care to model the junctions marked.)

Target/Response		
My baby.	<i>maɪ beɪbɪ</i>	2
Katy's party.	<i>dɛɪtɪθ baɪ..baʔdɪ</i>	0
Tommy cutting paper.	<i>tʊmɪ tʌtɪnd dʌ peɪpə</i>	0
Farmer sitting.	<i>fɑmɪ sɪʔsθɪn</i>	0
Daddy is talking. (/j/)	<i>dædɪ ʔɪç tʊʔtɪn</i>	0
A cup of tea. (C#V)	<i>ə tʌt pɒð ti</i>	0
Two sad cats. (C#C)	<i>tu sæd tæt</i>	1
Get an apple. (n)	<i>dɛt æd æpu</i>	0
Go in the house. (/w/)	<i>dəʊ ʔɪn də haʊθ</i>	0
Dinosaur museum.	(refused)	0
Do you like jelly? (intonation)	(refused)	0

Scoring: 2 = correct (local adult model)
1 = only 1 error

0 = more than 1 error/unable to attempt
NT= not tested

<i>Assimilation</i>	<i>Junction</i>
<i>Final consonant deletion</i>	<i>Rate of speech/stress patterns slow very careful production, equal stress</i>
<i>Cluster reduction</i>	<i>Pitch/voice quality</i>
<i>Imprecise vowels</i>	<i>Resonance</i>
<i>Imprecise consonants</i>	<i>Intonation</i>
<i>Syllable deletion</i>	<i>Compensatory strategies</i>

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